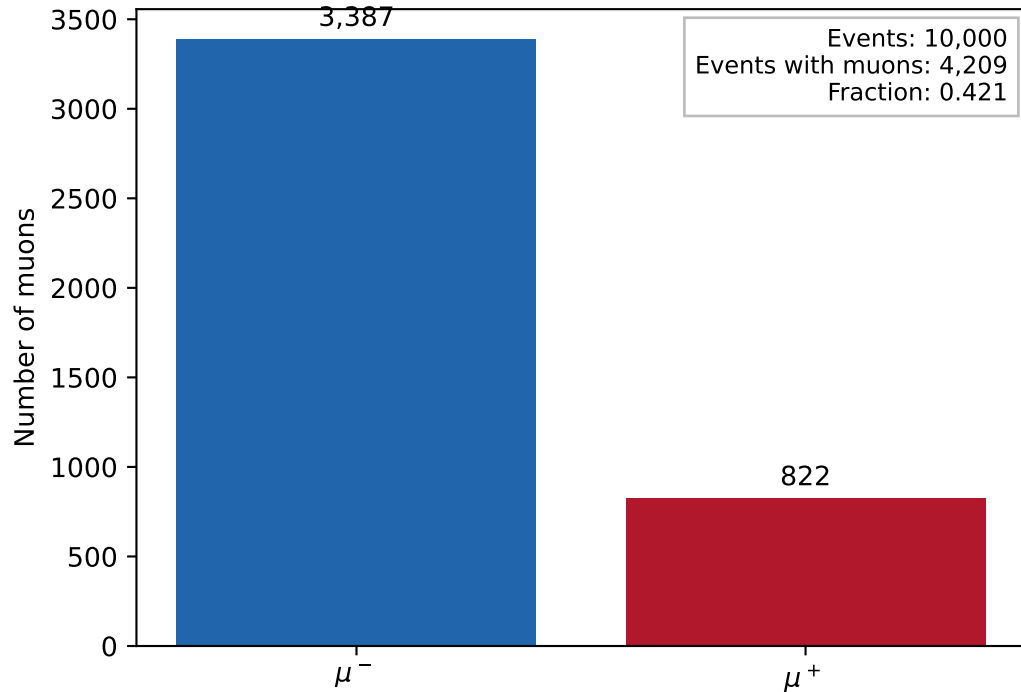
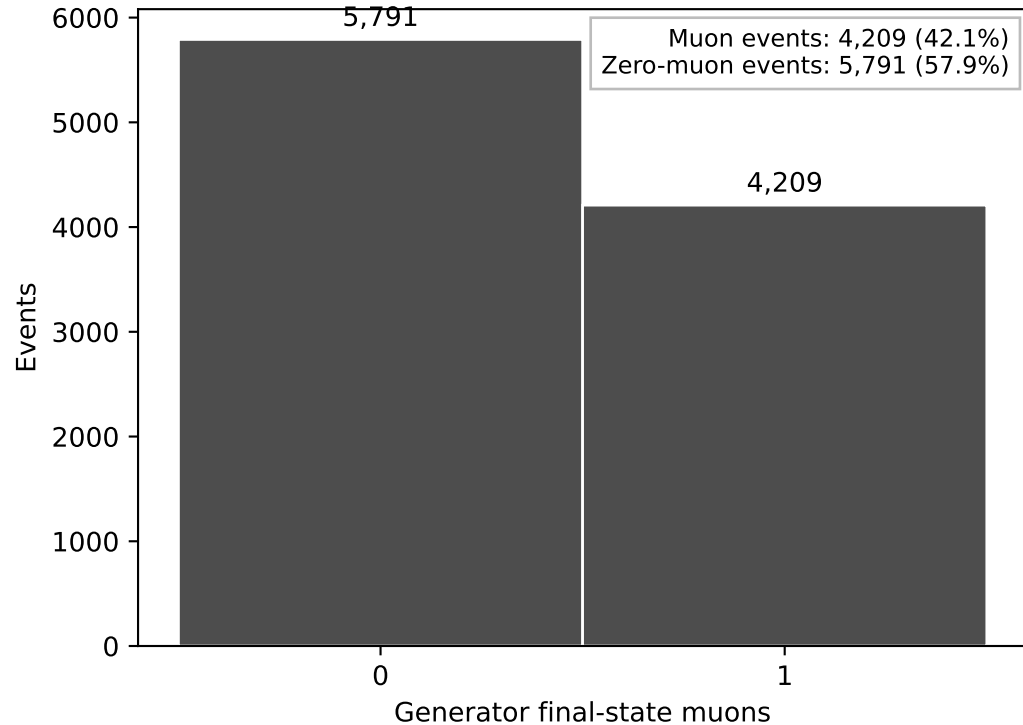


Atmospheric-neutrino muon truth overview

Generator final-state muon charges

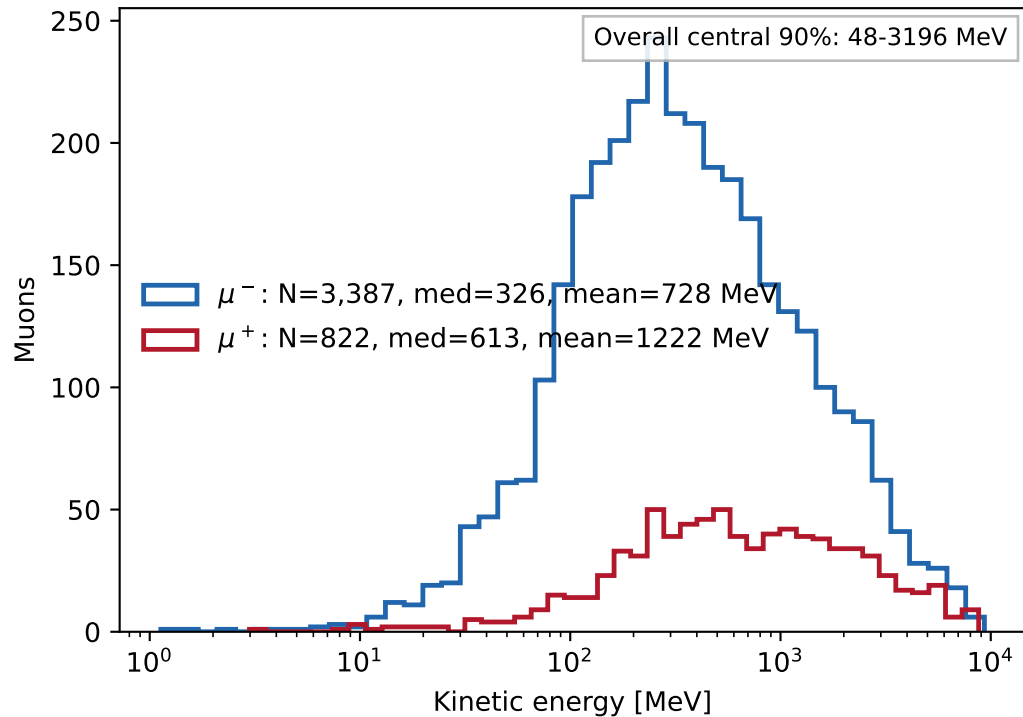


Muon multiplicity per event

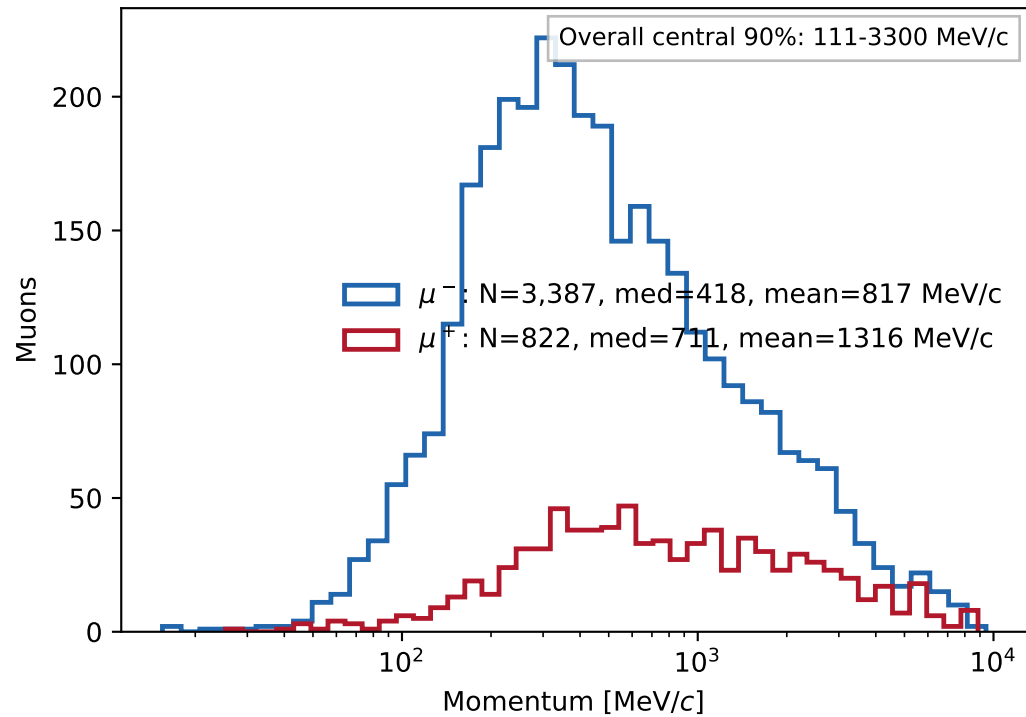


Generator final-state muon kinematics

Muon kinetic energy

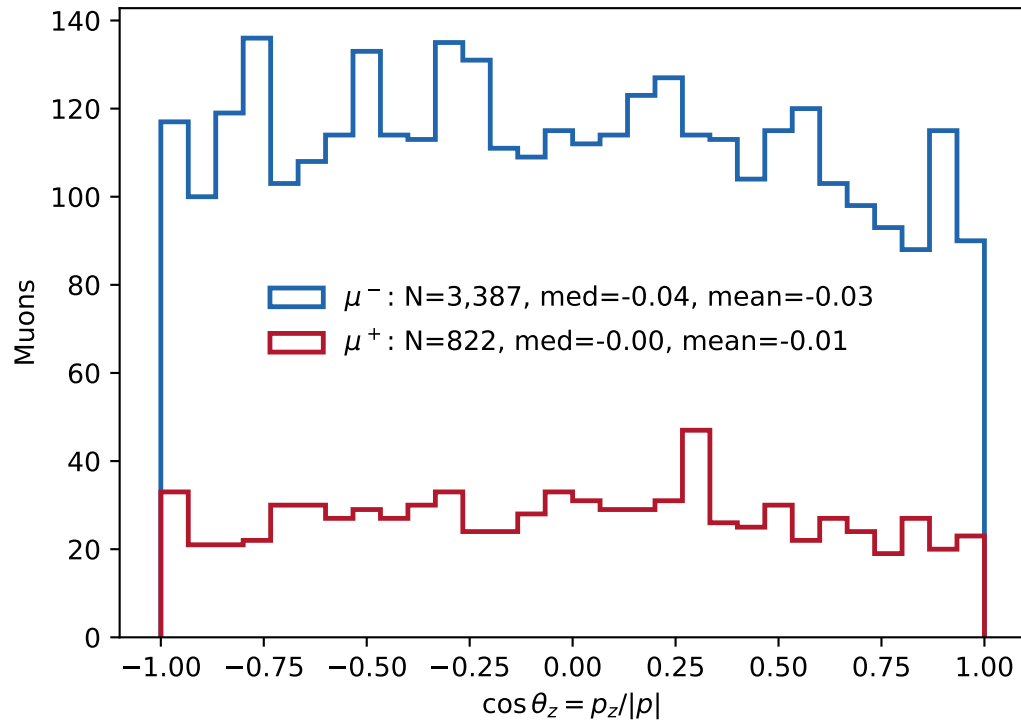


Muon momentum

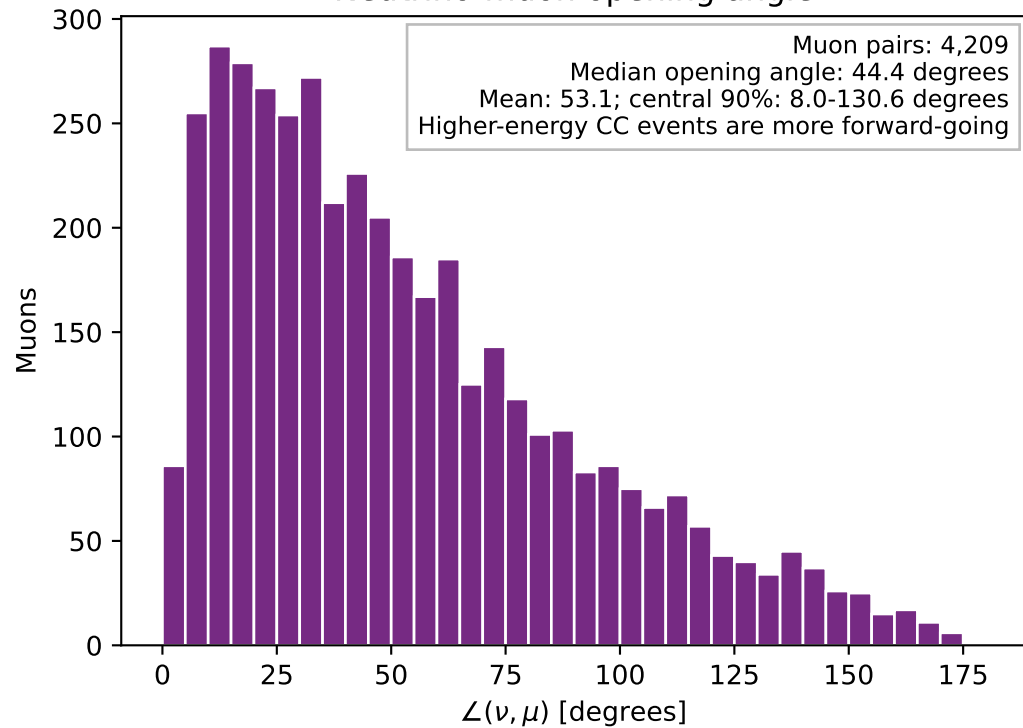


Muon angular distributions

Muon direction

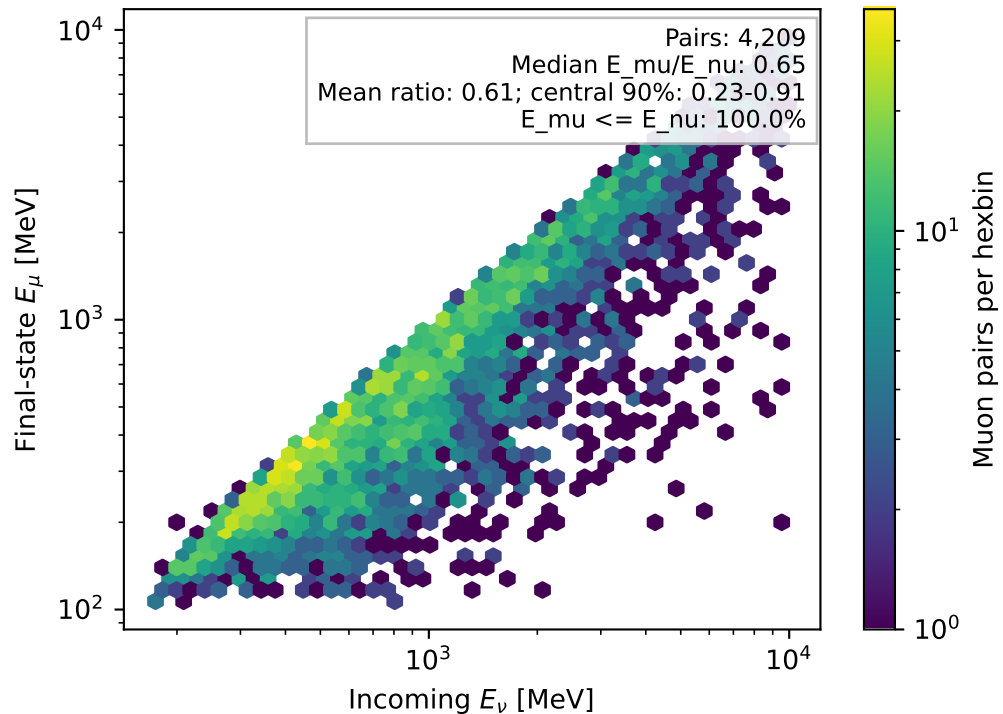


Neutrino-muon opening angle

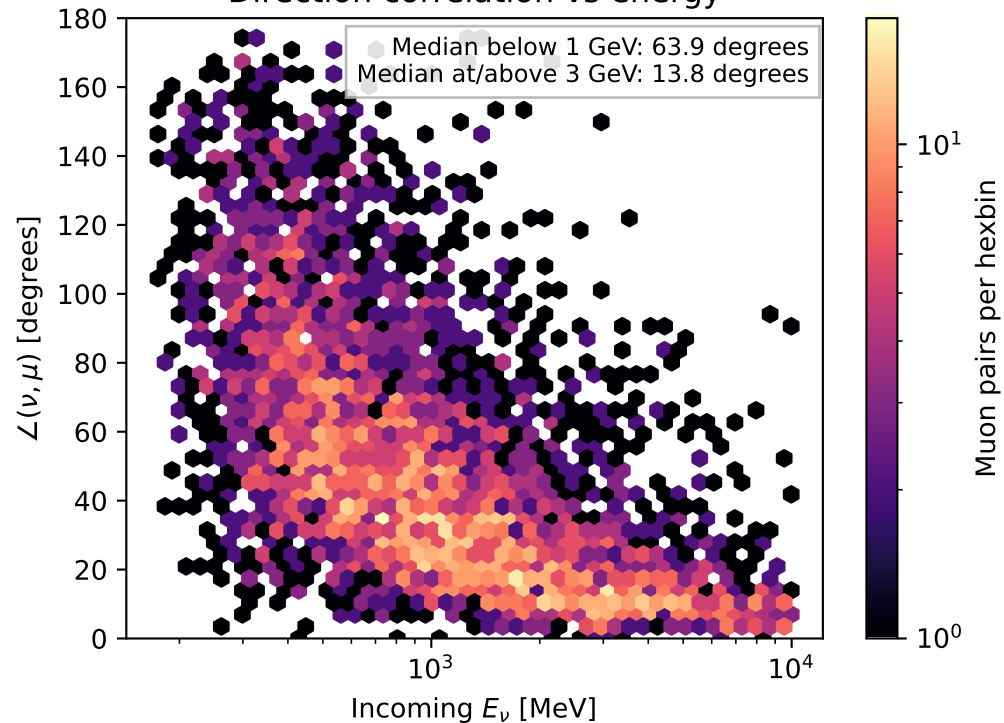


Neutrino-muon correlations

Incoming vs outgoing energy



Direction correlation vs energy



Muon sanity-check report

What 'generator final state' means

The generator initial state contains the incoming atmospheric neutrino and the argon target. GENIE then models the neutrino-argon interaction. The generator final state is the list of particles that emerge from that modeled interaction after GENIE's intranuclear final-state interactions and are handed to Geant4 for transport. The first-page bars therefore count muons produced by the neutrino interaction record, not detector hits and not every secondary muon later created inside Geant4.

Sample accounting

The sample contains 10,000 generated interactions. 4,209 events (42.1%) contain a generator final-state muon, with 3,387 μ^- and 822 μ^+ . Every selected event contains exactly one such muon. The μ^-/μ^+ ratio is 4.12.

Charge interpretation

4,209 of 4,209 selected muons (100.00%) have the charged-current sign expected from the incoming flavor: ν_μ produces μ^- , while $\bar{\nu}_\mu$ produces μ^+ . This strong correspondence is the central bookkeeping check. The larger μ^- population is physically reasonable because the generated atmospheric sample contains more neutrinos than antineutrinos and neutrino charged-current cross sections are generally larger at these energies.

Multiplicity interpretation

The multiplicity histogram has entries only at zero and one. That is sensible for the generator final state of this sample: an ordinary muon-flavor charged-current interaction supplies one primary charged lepton. Secondary muons from pion or kaon decays would belong to later Geant4 transport and are not included in this generator-final-state count.

Muon physics interpretation

Energy and momentum

The median muon kinetic energy is 363 MeV and the mean is 824 MeV; the central 90% spans 48 to 3196 MeV. The broad, right-skewed spectra are consistent with the broad atmospheric-neutrino spectrum. The mean exceeds the median because a relatively small high-energy tail pulls it upward, so the median is the more stable typical-event summary while the mean remains useful for energy-budget accounting. Muon total energy is below the incoming neutrino energy for 100.00% of pairs, and the median E_{μ}/E_{ν} is 0.65; the remaining energy goes into the hadronic system and nuclear recoil or removal energy.

Angular behavior

The overall median neutrino-muon opening angle is 44.4 degrees and the mean is 53.1 degrees. Below 1 GeV it is 63.9 degrees, while at and above 3 GeV it narrows to 13.8 degrees. This forward collimation with increasing energy is the expected charged-current behavior and is clearly visible in the correlation plot. It also shows why a low-energy muon is not a precise neutrino pointing proxy.

Zenith distribution

The muon $\cos(\theta_z)$ distribution is broadly populated across -1 to +1 without an obvious empty or singular region. That is qualitatively compatible with an all-sky atmospheric sample. A quantitative zenith validation requires confirming how the flux convention maps onto the simulation z axis and folding in oscillation weights; this plot alone should not be compared directly with a detector-level up-going/down-going spectrum.

Conclusion and limitations

The charge bookkeeping, one-muon multiplicity, energy ordering, and energy-dependent forward trend are all physically sensible. These plots validate internal kinematics and event handoff, but they do not by themselves validate an absolute atmospheric rate. Absolute comparisons require flux and oscillation weights, exposure, target normalization, and GENIE model-systematic variations.